What Should Economics Ask Next?

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Economics, machine learning, and metascience

Question choice is the scarce input.

- Published work leaves structured gaps.
- Some later papers add mechanisms.
- Others make direct links explicit.
- The graph helps decide what to inspect first.

Appendix

Question choice remains weakly formalized

- Choosing what to work on is one of the least formalized decisions in science.
- The stock of published work is large, fragmented, and hard to navigate.
- Researchers still face a conservative-search bias even when riskier moves matter.
- If AI lowers downstream costs, the bottleneck shifts further upstream toward question choice.

Jones (2009); Bloom et al. (2020); Foster, Rzhetsky, and Evans (2015)

Overview

This paper

We build and test

dated graph + walk-forward evaluation

- a dated graph from published economics-facing papers
- paper-local extraction, then cross-paper matching
- still-open questions are ranked at t and checked against later publication

We learn

substantive result

- the graph helps in two distinct historical families
- economics more often thickens mechanisms than later states locally implied direct links

Two dated question objects

Direct-to-path

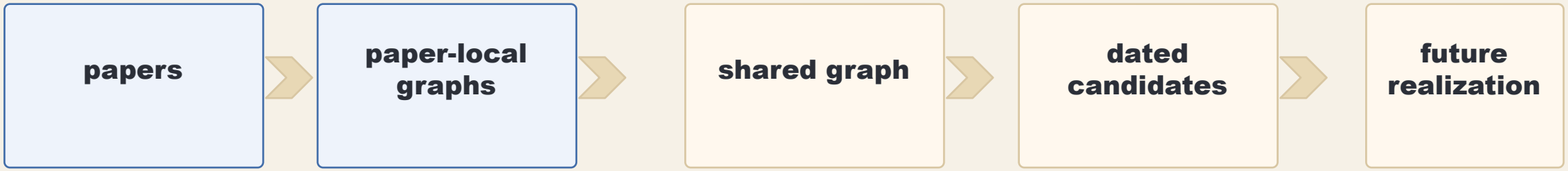
- the literature already contains a direct relation
- later work adds a clearer mediating path around it
- this is the mechanism-thickening object

Path-to-direct

- the literature already contains nearby support
- the direct relation itself is still missing
- later work states that direct link explicitly

Method bridge

From published papers to dated candidate questions

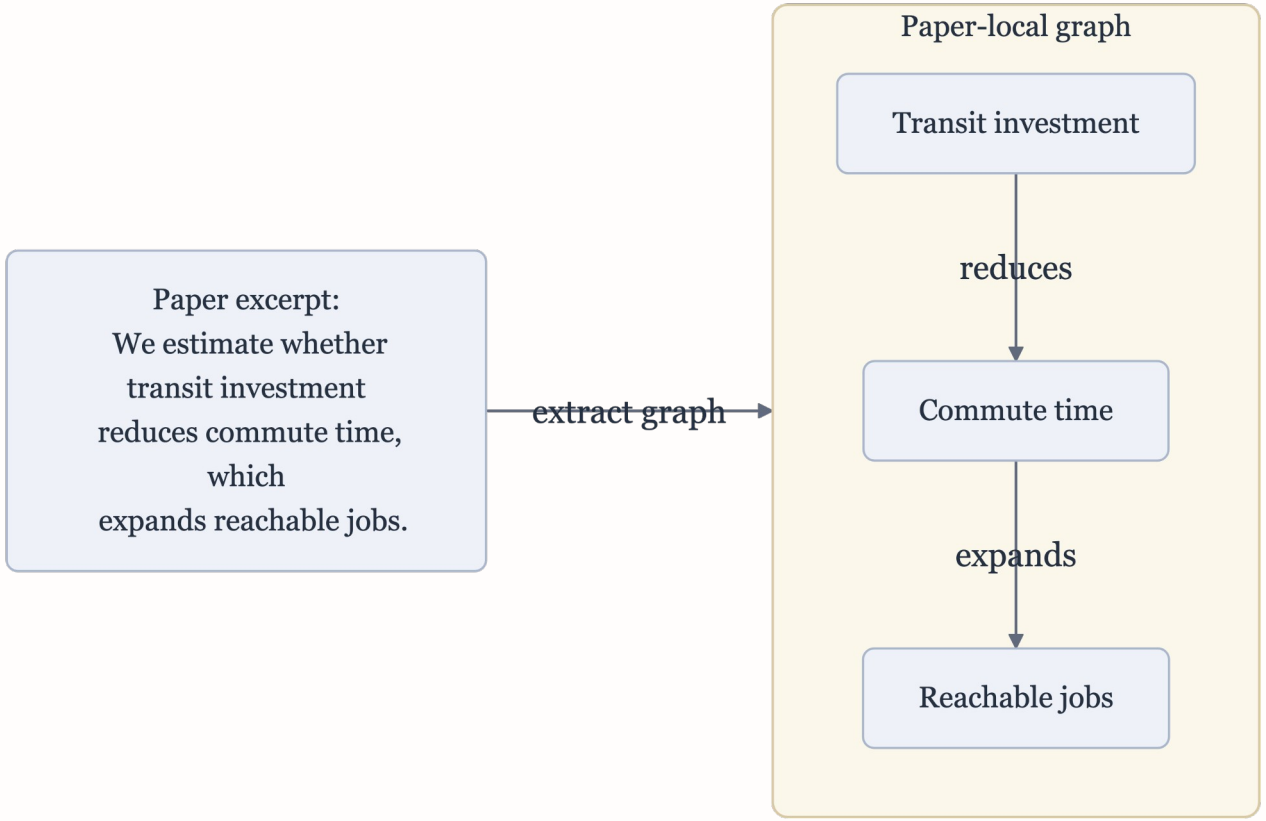


Freeze the literature at t-1, rank still-open directed questions, then check which later appear during [t, t+h].

Appendix

Each paper becomes a local graph

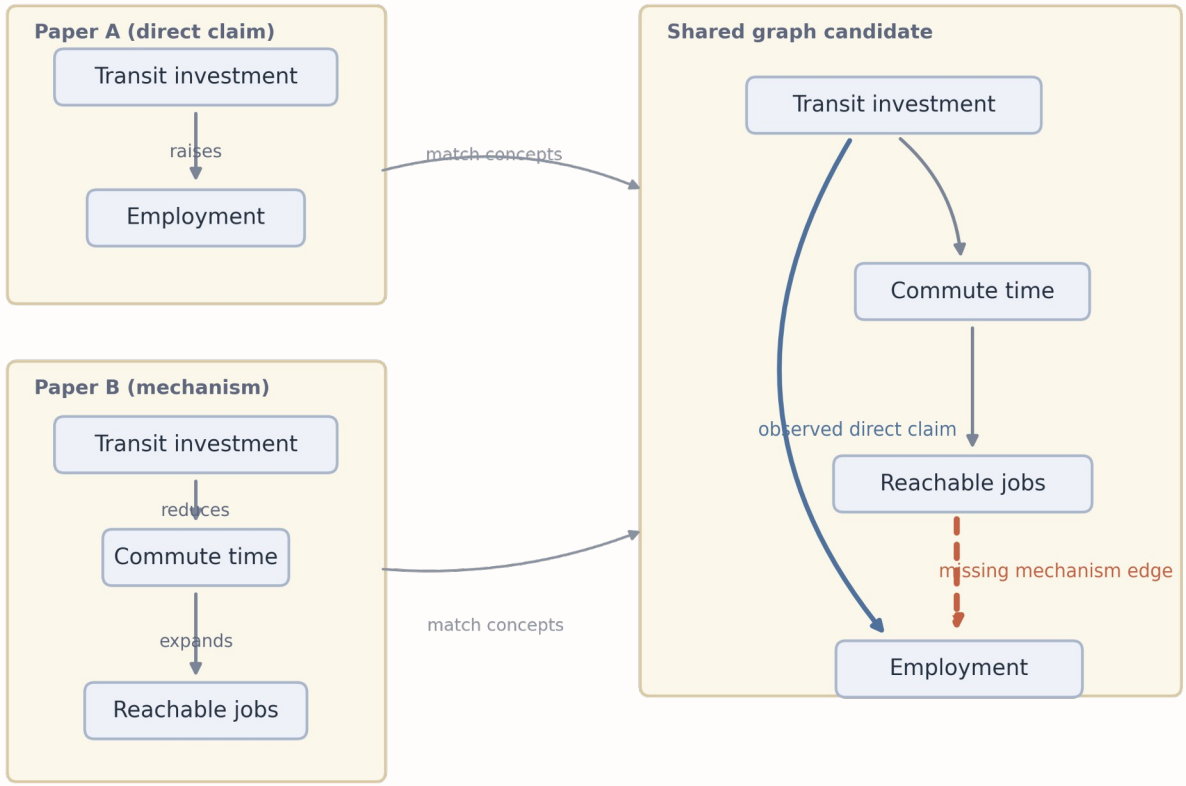
One excerpt -> one paper-local graph



Appendix

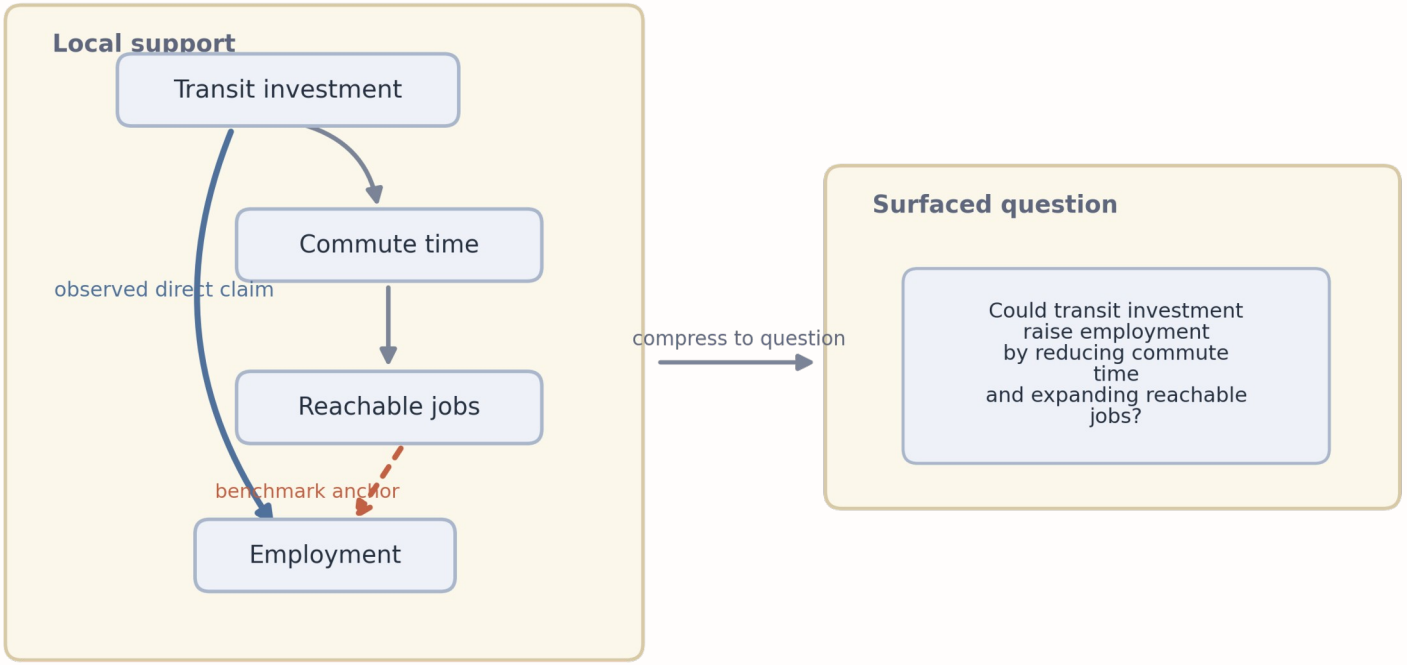
Candidate questions only exist after cross-paper matching

Two paper-local graphs -> one shared candidate neighborhood



The benchmark event is narrower than the surfaced question

What gets backtested versus what a reader would inspect

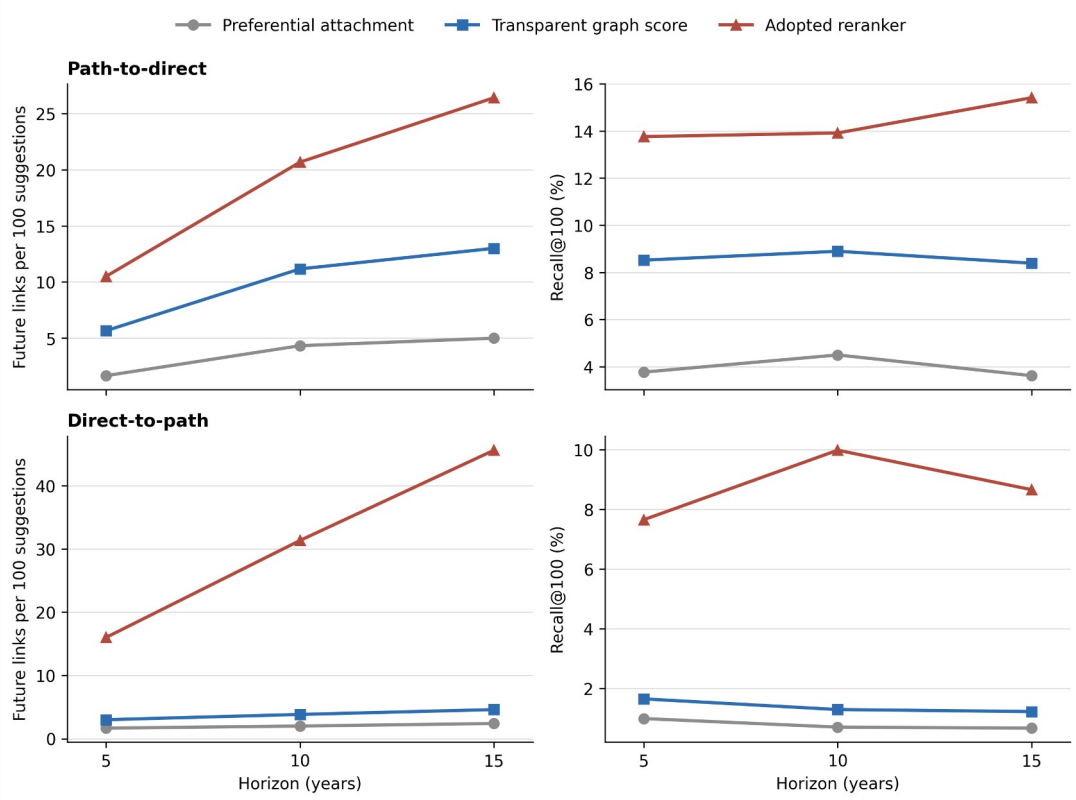


Historical result

Appendix

The graph improves on popularity in both families

Paired short-list benchmark



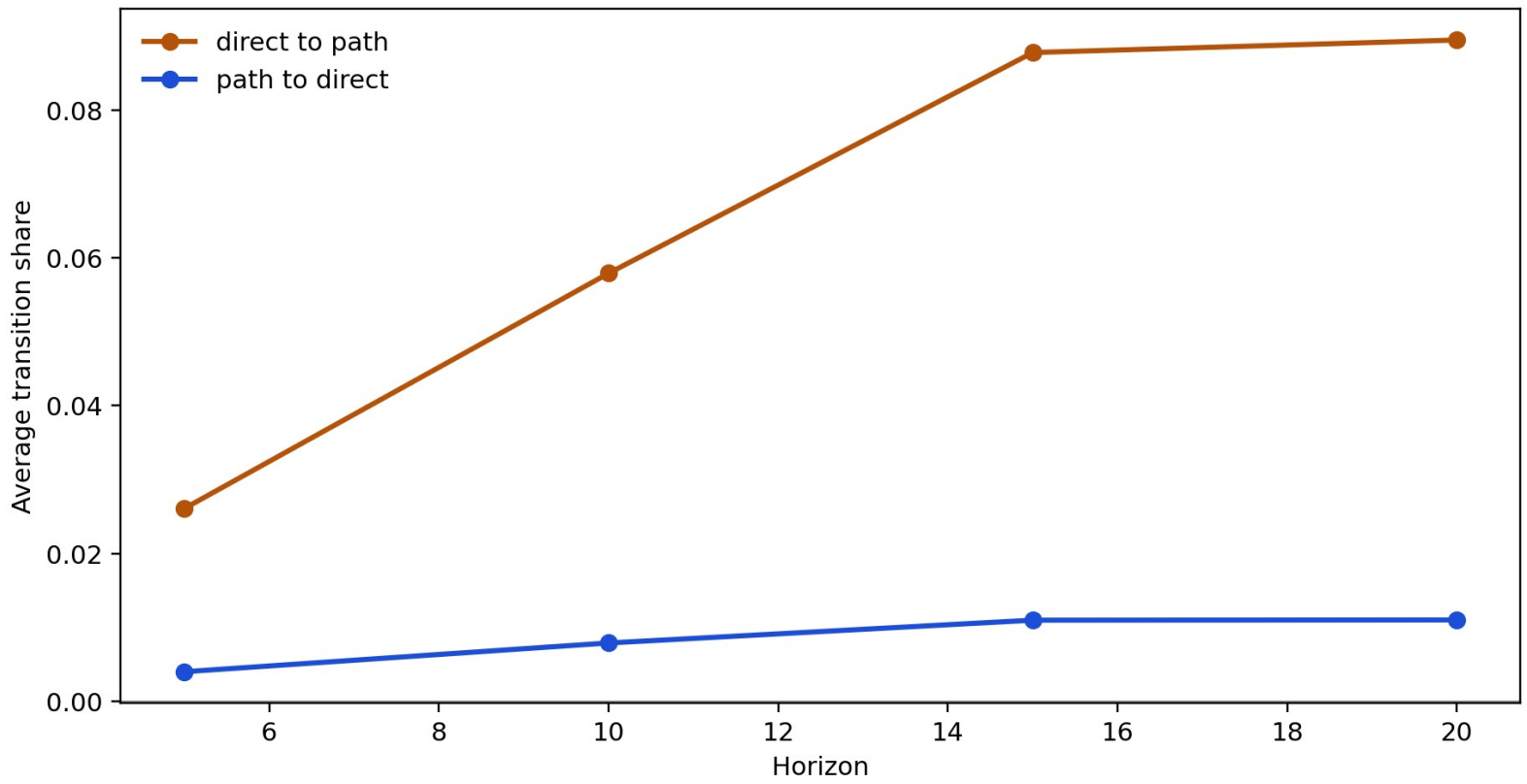
- Direct-to-path yields denser shortlists: more later realizations per 100 suggestions.
- Path-to-direct captures a larger share of all later-realized links; reranking improves both.

Substantive result

Appendix

Mechanism thickening is more common than direct closure

How the two forms of progress compare



The literature more often deepens mechanisms around existing claims than later closes locally implied direct links.

Current frontier

What the graph surfaces now

Appendix

Direct-to-path

known relation, missing mechanism

- Could digital financial inclusion support employment through labour matching or market access?
- Could tax refunds lift productivity through liquidity and investment?
- Could nutrition information shift willingness to pay through health-salience channels?

Path-to-direct

nearby support, missing direct relation

- Could trade liberalization raise R&D?
- Could the digital economy strengthen environmental regulation?
- Could renewable energy reshape urbanization?

Appendix

What we built and learned

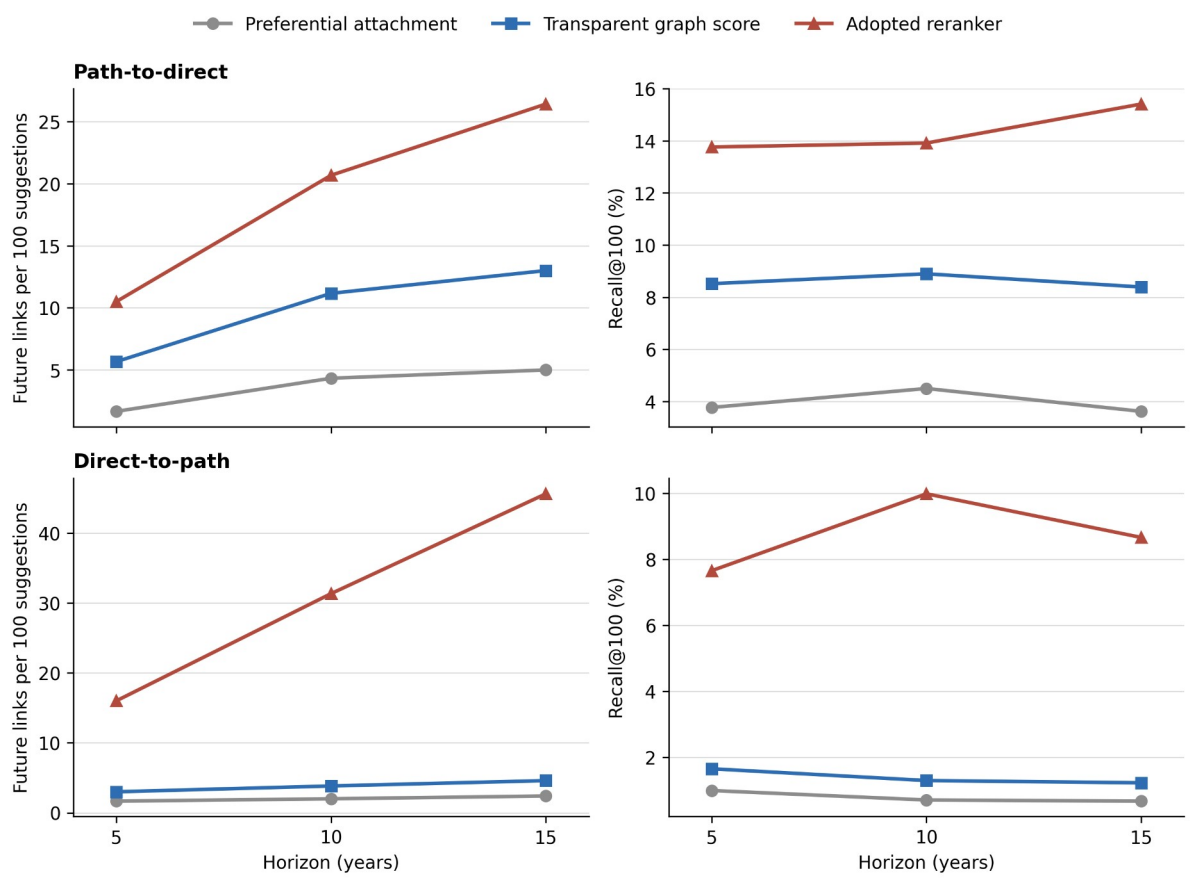
- built a dated graph from published economics-facing papers
- tested two benchmarkable forms of still-open next questions
- showed that graph structure helps when reading time is scarce
- found that the literature more often thickens mechanisms than it later states locally implied direct links

Thank you

Benchmark detail

Back to main

Paired benchmark result



Appendix detail: paired main benchmark.

Method detail

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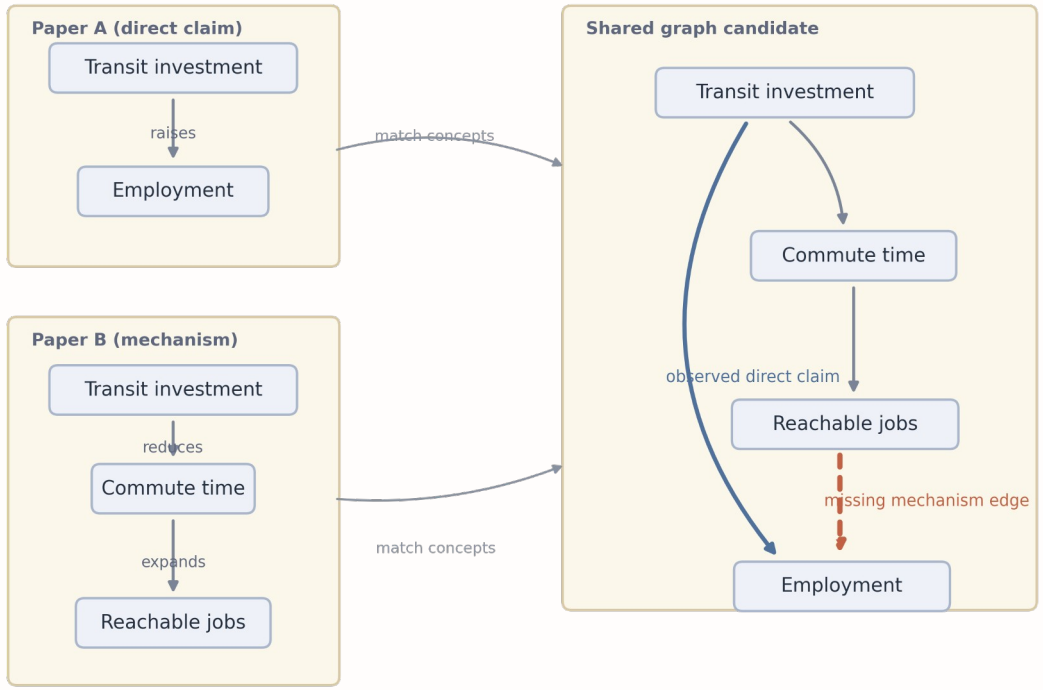
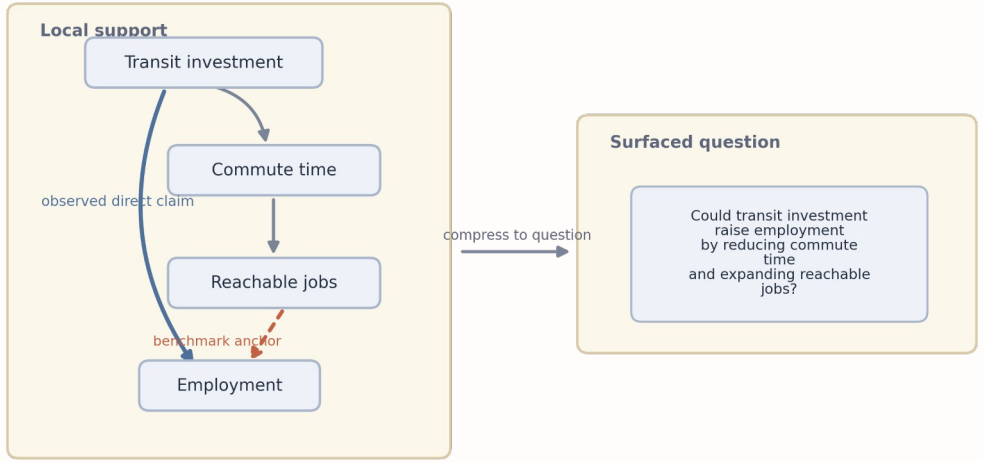
Extraction and matching



Benchmark anchor versus surfaced question



Appendix detail: methodology figures.

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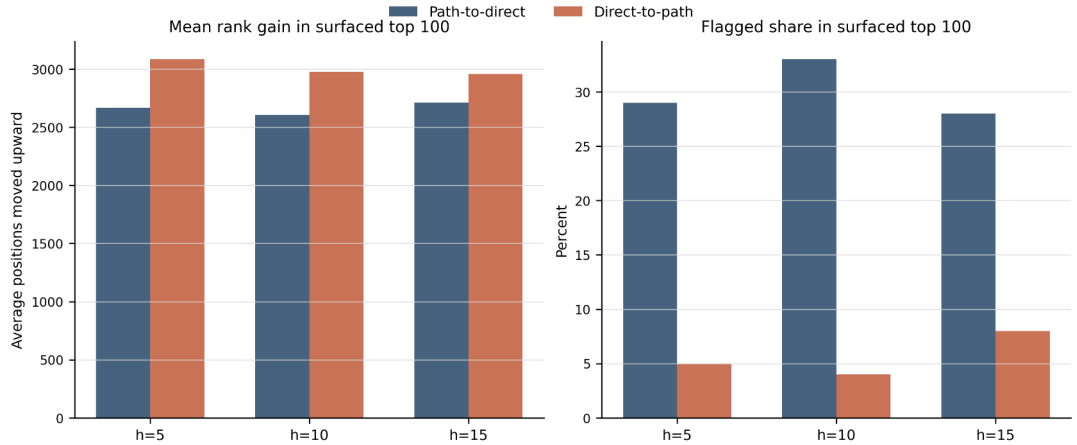
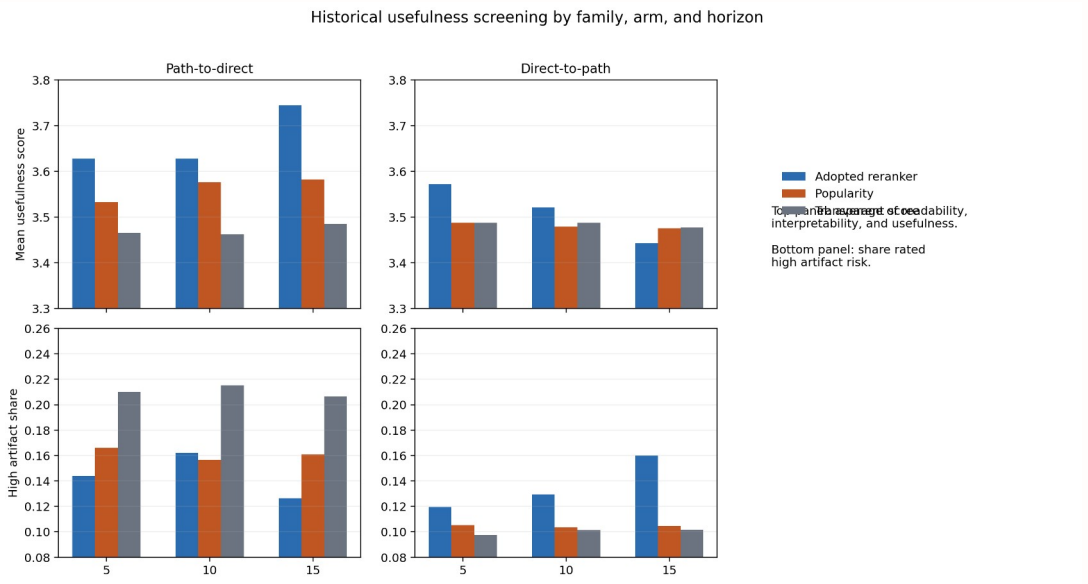
Extra empirical views

Current frontier summary



Paired usefulness screen



Appendix detail: surfaced examples and workflow screen.